



Atoms, Elements & Compounds

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Comprehensive study of atomic structure, subatomic particles, isotopes and relative atomic mass calculations, electronic configuration, the periodic table, ionic bonding and giant ionic lattices, simple covalent molecules and giant macromolecular structures (diamond, graphite, silicon dioxide), and metallic bonding with alloys.



1. Elements, Compounds & Mixtures

Extended tier only

Fundamental Definitions of Matter

All matter can be classified into elements, compounds, or mixtures depending on atomic composition and chemical bonding:

1. **Element:** A pure substance composed of only **one type of atom**. Elements cannot be broken down into simpler chemical substances by any chemical reaction (e.g. Iron, Fe; Oxygen gas, O₂; Copper, Cu).
2. **Compound:** A pure substance formed when **two or more different elements chemically combine** in a **fixed, definite ratio** held together by chemical bonds (e.g. Water, H₂O; Sodium chloride, NaCl; Carbon dioxide, CO₂).
3. **Mixture:** A physical combination of **two or more substances** (elements or compounds) that are **not chemically joined**. The components can be present in **any proportion** and retain their individual physical and chemical properties (e.g. Air, crude oil, brass, sea water).

Comparison: Compounds vs Mixtures

Feature	Compound	Mixture
Composition	Elements combined in a fixed, definite ratio	Components mixed in any variable ratio
Chemical Bonding	Formed by chemical bonds (ionic or covalent)	No chemical bonds formed between components
Separation Method	Can only be separated by chemical reactions (or electrolysis)	Easily separated by physical separation techniques (filtration, distillation, chromatography)
Properties	Completely new properties different from constituent elements	Retains the individual properties of each component
Energy Change on Formation	Usually accompanied by significant energy transfer (exothermic or endothermic)	Little or no energy change occurs during mixing

Examiner Tips and Tricks: Classic Cambridge Exam Comparison: Iron & Sulfur

Mixture of Fe and S:

Grey iron filings and yellow sulfur powder physically mixed.

A magnet attracts and extracts the iron filings completely (physical separation).

Dilute hydrochloric acid added to the mixture releases hydrogen gas (H₂) from the iron.

Compound: Iron(II) Sulfide (FeS):

Formed by heating the mixture strongly until an exothermic glow persists.

The resulting black solid is **not attracted to a magnet**.

Reaction with dilute hydrochloric acid releases toxic, foul-smelling hydrogen sulfide gas (H₂S), proving an entirely new substance was formed.

Physical Changes vs Chemical Changes

- **Physical Change:** A change in state, shape, or particle arrangement where **no new chemical substances are formed**. Physical changes are easily **reversible** by physical means (e.g. melting ice, dissolving sodium chloride in water, boiling ethanol).

- **Chemical Change:** A process in which chemical bonds are broken and new bonds are formed, resulting in **one or more new chemical substances**. Chemical changes are usually **irreversible** or difficult to reverse (e.g. combustion of methane, rusting of iron, thermal decomposition of calcium carbonate).

- **Observable signs of chemical reactions:**

1. Evolution of a gas (effervescence / bubbling).
2. Formation of an insoluble solid (a precipitate).
3. Significant change in temperature (exothermic heat release or endothermic cooling).
4. Distinct color change.

2. Atomic Structure & The Periodic Table

Extended tier only

The Structure of the Atom

An atom consists of a tiny, extremely dense **central nucleus** surrounded by electrons orbiting in concentric energy levels (**electron shells**):

1. **The Nucleus:** Located at the center of the atom, containing **protons** and **neutrons** (collectively termed **nucleons**). The nucleus carries a positive charge and accounts for almost all the mass of the atom.
2. **Electron Shells:** Electrons orbit the nucleus at distinct distances. Most of an atom's volume is empty space.

Properties of Subatomic Particles

Particle	Location	Relative Mass	Relative Charge	Symbol
Proton	In the nucleus	1	+1	p
Neutron	In the nucleus	1	0 (neutral)	n
Electron	In electron shells around the nucleus	$\frac{1}{1840}$ (or 0.0005, negligible)	-1	e ⁻

Key Examiner Notes: Atomic Neutrality

Atoms are electrically neutral overall because the **number of positively charged protons equals the number of negatively charged electrons**:

$$\text{Net Charge} = (\text{Number of Protons} \times (+1)) + (\text{Number of Electrons} \times (-1)) = 0$$

Atomic Number (*Z*) and Mass Number (*A*)

In standard chemical nuclide notation:



- **Proton Number (Atomic Number, *Z*):** The number of protons in the nucleus of an atom. The proton number uniquely identifies the chemical element.
- **Mass Number (Nucleon Number, *A*):** The total number of protons and neutrons in the nucleus of an atom:

$$A = \text{Number of Protons} + \text{Number of Neutrons}$$

- **Deducing Subatomic Particle Counts:**
- Number of Protons = *Z*
- Number of Electrons (in neutral atom) = *Z*
- Number of Neutrons = *A* - *Z*

Key Examiner Note

Worked Example: For sodium, $^{23}_{11}\text{Na}$:

- Protons = 11
- Electrons = 11
- Neutrons = $23 - 11 = 12$

⚡ Electronic Configuration (Elements 1 to 20)

Electrons occupy concentric electron shells around the nucleus according to specific rules:

1. **Shell 1** (closest to nucleus): Holds a maximum of 2 **electrons**.
2. **Shell 2**: Holds a maximum of 8 **electrons**.
3. **Shell 3**: Holds a maximum of 8 **electrons** (for the first 20 elements).
4. **Shell 4**: Accommodates remaining electrons (up to 2 for Calcium, $Z = 20$).

Electron Configurations of the First 20 Elements:

Element	Symbol	Proton No. (Z)	Electron Configuration	Group	Period
Hydrogen	H	1	1	1	1
Helium	He	2	2 (Duplet)	VIII/0	1
Lithium	Li	3	2, 1	I	2
Beryllium	Be	4	2, 2	II	2
Boron	B	5	2, 3	III	2
Carbon	C	6	2, 4	IV	2
Nitrogen	N	7	2, 5	V	2
Oxygen	O	8	2, 6	VI	2
Fluorine	F	9	2, 7	VII	2
Neon	Ne	10	2, 8 (Octet)	VIII/0	2
Sodium	Na	11	2, 8, 1	I	3
Magnesium	Mg	12	2, 8, 2	II	3
Aluminium	Al	13	2, 8, 3	III	3
Silicon	Si	14	2, 8, 4	IV	3
Phosphorus	P	15	2, 8, 5	V	3
Sulfur	S	16	2, 8, 6	VI	3
Chlorine	Cl	17	2, 8, 7	VII	3
Argon	Ar	18	2, 8, 8 (Octet)	VIII/0	3
Potassium	K	19	2, 8, 8, 1	I	4
Calcium	Ca	20	2, 8, 8, 2	II	4

Relationship with the Periodic Table

1. **Group Number:** Corresponds to the **number of electrons in the outer shell** (valence electrons).
 - Group I elements have 1 outer electron.
 - Group VII elements have 7 outer electrons.
 - Elements in the same group have similar chemical properties because they have the same number of outer-shell electrons.
2. **Period Number:** Corresponds to the **number of occupied electron shells**.
 - Period 2 elements have 2 occupied shells.
 - Period 3 elements have 3 occupied shells.
3. **Noble Gases (Group VIII / 0):** Possess a **full outer shell** of electrons (a stable duplet for He with 2, and a stable octet of 8 for Ne, Ar). This electronic configuration makes them **chemically inert** (unreactive).

3. Isotopes & Relative Atomic Mass

Extended tier only

Definition of Isotopes

Isotopes are different atoms of the **same element** that have:

- The **same number of protons** (same atomic / proton number).
- A **different number of neutrons** (different mass / nucleon number).

Common Examples of Isotopes:

- **Hydrogen:**
 - Protium: ${}^1_1\text{H}$ (1 proton, 0 neutrons, 1 electron)
 - Deuterium: ${}^2_1\text{H}$ (1 proton, 1 neutron, 1 electron)
 - Tritium: ${}^3_1\text{H}$ (1 proton, 2 neutrons, 1 electron)
- **Carbon:**
 - Carbon-12: ${}^{12}_6\text{C}$ (6 protons, 6 neutrons)
 - Carbon-13: ${}^{13}_6\text{C}$ (6 protons, 7 neutrons)
 - Carbon-14: ${}^{14}_6\text{C}$ (6 protons, 8 neutrons)
- **Chlorine:**
 - Chlorine-35: ${}^{35}_{17}\text{Cl}$ (17 protons, 18 neutrons)
 - Chlorine-37: ${}^{37}_{17}\text{Cl}$ (17 protons, 20 neutrons)

Chemical vs Physical Properties of Isotopes

Key Examiner Notes & Exam Priorities: Core Cambridge Exam Concept: Chemical Reactivity of Isotopes

Chemical properties are identical:

Chemical reactions depend entirely on the **number and arrangement of valence (outer shell) electrons**.

Because isotopes of the same element have the **exact same number of electrons** and the same electronic configuration, they undergo identical chemical reactions.

Physical properties differ slightly:

Physical properties such as **density, melting point, boiling point**, and **rate of diffusion** depend on atomic mass.

Heavier isotopes have slightly higher densities, higher boiling points, and diffuse more slowly than lighter isotopes.

Calculating Relative Atomic Mass (A_r)

The **Relative Atomic Mass (A_r)** of an element is the **weighted average mass** of naturally occurring isotopes of that element relative to $\frac{1}{12}$ of the mass of an atom of carbon-12.

Formula:

$$A_r = \frac{\sum(\text{Isotopic Mass} \times \text{Percentage Abundance})}{100}$$

$$A_r = \frac{(m_1 \times \%_1) + (m_2 \times \%_2) + \dots}{100}$$

Step-by-Step Worked Calculations

Example 1: Chlorine

A sample of naturally occurring chlorine consists of:

- 75% of ^{35}Cl
- 25% of ^{37}Cl

$$A_r(\text{Cl}) = \frac{(35 \times 75) + (37 \times 25)}{100} = \frac{2625 + 925}{100} = \frac{3550}{100} = 35.5$$

Example 2: Magnesium (3 Isotopes)

A sample of magnesium contains three isotopes:

- ^{24}Mg (abundance 79%)
- ^{25}Mg (abundance 10%)
- ^{26}Mg (abundance 11%)

$$A_r(\text{Mg}) = \frac{(24 \times 79) + (25 \times 10) + (26 \times 11)}{100} = \frac{1896 + 250 + 286}{100} = \frac{2432}{100} = 24.32$$

Radioactive vs Non-Radioactive Isotopes

- **Non-radioactive isotopes:** Stable nuclei that do not decay over time (e.g. ^{12}C , ^{16}O).
- **Radioactive isotopes (Radioisotopes):** Unstable nuclei that spontaneously decay, emitting radiation (α , β , γ rays).
- **Medical uses:** Cobalt-60 is used in radiation therapy to treat cancer cells; Iodine-131 is used as a medical tracer to monitor thyroid gland activity.
- **Industrial uses:** Carbon-14 is used in archaeological radiocarbon dating; Uranium-235 is used as nuclear fuel in power generation.

4. Ions & Ionic Bonding

Extended tier only

⚡ Formation of Ions: Electron Transfer

Atoms undergo chemical reactions to achieve a **stable electronic configuration** with a full outer shell of electrons (like the noble gases):

1. Metal Atoms (Loss of Electrons):

- Metals have 1, 2, or 3 valence electrons.
- They **lose** their outer shell electrons to form positively charged ions called **cations**.
- Example: A neutral sodium atom (2, 8, 1) loses 1 electron to form a sodium ion Na^+ (2, 8):



- Example: A magnesium atom (2, 8, 2) loses 2 electrons to form Mg^{2+} (2, 8):



2. Non-Metal Atoms (Gain of Electrons):

- Non-metals have 5, 6, or 7 valence electrons.
- They **gain** electrons into their outer shell to form negatively charged ions called **anions** (ending in *-ide*).
- Example: A neutral chlorine atom (2, 8, 7) gains 1 electron to form a chloride ion Cl^- (2, 8, 8):



- Example: An oxygen atom (2, 6) gains 2 electrons to form an oxide ion O^{2-} (2, 8):



📄 Common Ions and Their Valencies

Charge	Common Cations (Metals)	Common Anions (Non-metals & Compound Ions)
+1 / -1	Li^+ , Na^+ , K^+ , Ag^+ , H^+ , NH_4^+ (Ammonium)	F^- , Cl^- , Br^- , I^- , OH^- (Hydroxide), NO_3^- (Nitrate)
+2 / -2	Mg^{2+} , Ca^{2+} , Ba^{2+} , Cu^{2+} , Fe^{2+} , Zn^{2+} , Pb^{2+}	O^{2-} , S^{2-} , SO_4^{2-} (Sulfate), CO_3^{2-} (Carbonate)
+3 / -3	Al^{3+} , Fe^{3+}	N^{3-} , PO_4^{3-} (Phosphate)

🔗 What is an Ionic Bond?

🔑 Key Examiner Notes: 📖 Cambridge Syllabus Definition

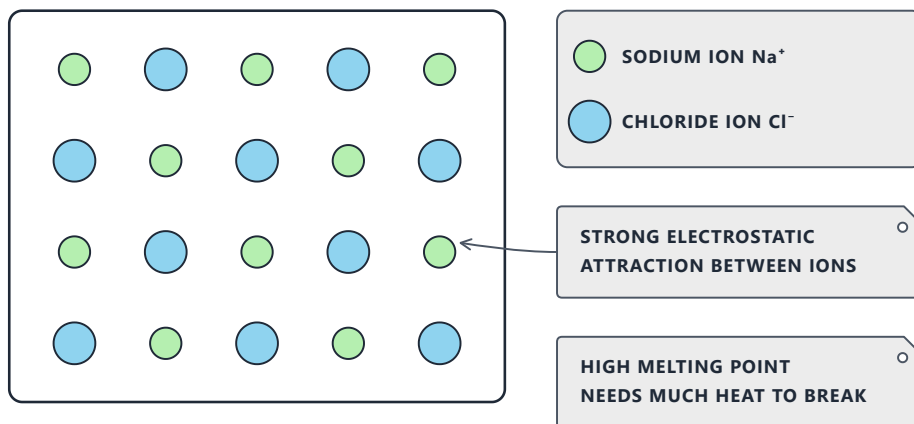
An **ionic bond** is the **strong electrostatic attraction between oppositely charged ions** (cations and anions).

Ionic bonds are formed when electrons are transferred from a metal atom to a non-metal atom:

- In sodium chloride (NaCl), one electron is transferred from Na to Cl, forming Na^+ and Cl^- .
- In magnesium oxide (MgO), two electrons are transferred from Mg to O, forming Mg^{2+} and O^{2-} .
- In calcium chloride (CaCl_2), calcium transfers 1 electron to each of two chlorine atoms, forming Ca^{2+} and $2 \times \text{Cl}^-$.

The Giant Ionic Lattice Structure

Ionic compounds do not exist as isolated single molecules. Instead, they form a **giant three-dimensional ionic lattice** consisting of billions of alternating positive and negative ions held together by electrostatic attractions acting in all directions.



Physical Properties of Ionic Compounds Explained

Physical Property	Observation	Explanation at Particle Level
Melting & Boiling Points	Very High (e.g. NaCl m.p. is 801°C ; MgO is 2852°C)	The giant lattice is held by many strong electrostatic attractions between oppositely charged ions in all dimensions. Breaking these bonds requires a large amount of thermal energy .
Electrical Conductivity (Solid)	Poor conductor / Insulator	Ions are locked tightly into fixed positions within the regular lattice and cannot move to carry electric charge.
Electrical Conductivity (Molten/Aqueous)	Good electrical conductor	When melted or dissolved in water, the crystal lattice breaks down. The ions become free to move and migrate towards electrodes, carrying electrical current.
Solubility	Usually soluble in water , insoluble in non-polar organic solvents	Polar water molecules interact strongly with the charged ions, hydrating them and pulling them away from the lattice.
Brittleness	Brittle (shatters when struck with a hammer)	When a mechanical force is applied, layers of ions slide over each other. This brings ions of like charge into direct alignment (e.g. Na^+ next to Na^+), causing severe electrostatic repulsion that splits the crystal along cleavage planes.



Examiner Warning & Common Misconceptions: Crucial Examiner Report Trap: Ionic Conduction

Candidates frequently lose marks by writing: *"Sodium chloride conducts electricity when molten because electrons are free to move."*

THIS IS INCORRECT AND SCORES ZERO MARKS.

Metals conduct because of mobile **electrons**.

Molten ionic compounds conduct because **IONS are free to move**, NOT electrons! Always specify **mobile ions**.

5. Simple Molecules & Covalent Bonding

Extended tier only

What is a Covalent Bond?

Key Examiner Notes: Cambridge Syllabus Definition

A **covalent bond** is formed when a **pair of electrons is shared between two non-metal atoms**.

The shared pair of electrons is held in place by the **electrostatic attraction between the positive nuclei of both atoms and the shared negative electrons**.

Non-metal atoms share electrons to achieve a complete outer energy level (noble gas structure):

- **Single Covalent Bond:** 1 shared pair of electrons (2 electrons shared, e.g. $\text{H}-\text{H}$, $\text{Cl}-\text{Cl}$, $\text{H}-\text{Cl}$).
- **Double Covalent Bond:** 2 shared pairs of electrons (4 electrons shared, e.g. $\text{O}=\text{O}$, $\text{O}=\text{C}=\text{O}$).
- **Triple Covalent Bond:** 3 shared pairs of electrons (6 electrons shared, e.g. $\text{N}\equiv\text{N}$).

Dot-and-Cross Diagrams Required by Cambridge IGCSE (0620)

Candidates must be able to draw or deduce dot-and-cross representations showing outer shell electrons for the following key molecules:

1. Diatomic Molecules

- **Hydrogen (H_2):**
 - Each H atom shares 1 electron to form 1 single bond.
 - Both atoms achieve a stable duplet (2 electrons in shell 1): $\text{H}-\text{H}$.
- **Chlorine (Cl_2):**
 - Each Cl atom (2, 8, 7) contributes 1 electron to form 1 shared pair.
 - Each chlorine atom retains 3 non-bonding lone pairs (6 non-bonding electrons).
- **Oxygen (O_2):**
 - Each O atom (2, 6) needs 2 electrons; shares 2 pairs of electrons (4 shared electrons in total).
 - Forms a **double covalent bond** ($\text{O}=\text{O}$); each oxygen has 2 lone pairs.
- **Nitrogen (N_2):**
 - Each N atom (2, 5) needs 3 electrons; shares 3 pairs of electrons (6 shared electrons).
 - Forms a **triple covalent bond** ($\text{N}\equiv\text{N}$); each nitrogen has 1 lone pair.

2. Inorganic Hydrides & Halides

- **Hydrogen Chloride (HCl):**
 - 1 shared pair between H and Cl. Chlorine retains 3 lone pairs.
- **Water (H_2O):**
 - Central oxygen atom shares 1 electron pair with each of two hydrogen atoms (2 single bonds).
 - Oxygen retains 2 **unshared lone pairs** of electrons, giving water a bent (V-shaped) geometry.
- **Ammonia (NH_3):**
 - Central nitrogen atom forms 3 single covalent bonds with three hydrogen atoms.
 - Nitrogen retains 1 **unshared lone pair** of electrons, giving ammonia a trigonal pyramidal shape.

3. Organic & Carbon Molecules

- **Methane (CH_4):**
 - Central carbon atom (2, 4) shares each of its 4 outer electrons with a hydrogen atom (4 single C-H bonds).

- All 8 outer valence positions of carbon are involved in bonding (tetrahedral shape).
- **Carbon Dioxide (CO₂):**
- Central carbon atom forms **two double covalent bonds** with two oxygen atoms: O=C=O.
- 4 shared pairs in total (8 shared electrons). Each oxygen retains 2 lone pairs.
- **Ethene (C₂H₄):**
- Double bond between carbon atoms (C=C, 2 shared pairs) and four single C–H bonds.
- **Methanol (CH₃OH):**
- Carbon bonded to 3 hydrogens and 1 oxygen; oxygen bonded to 1 hydrogen and holding 2 lone pairs.

🌐 Structure & Physical Properties of Simple Molecular Substances

Substances consisting of simple covalent molecules have distinct physical characteristics:

Property	Observation	Particle Explanation
Melting & Boiling Points	Low (many are gases or liquids at room temperature, e.g. O ₂ , CH ₄ , H ₂ O)	Within each molecule, the covalent bonds are very strong . However, between separate molecules there are only weak intermolecular forces . Very little thermal energy is required to overcome these weak intermolecular forces during melting or boiling.
Electrical Conductivity	Non-conductors (Insulators) in all states	Simple molecules are neutral and have no overall electric charge . There are no mobile ions and no free delocalised electrons to move and carry charge.
Volatility	High (evaporate readily)	Because intermolecular forces are weak, molecules at the liquid surface easily gain enough energy to escape into the gas phase.
Solubility	Generally insoluble in water , soluble in non-polar organic solvents	Most simple covalent molecules cannot form hydrogen bonds or electrostatic interactions with polar water molecules.

🔴 Examiner Warning & Common Misconceptions: Cambridge Examiner Warning: "Breaking Covalent Bonds"

When explaining why water boils at 100°C or why methane melts at –182°C:

DO NOT WRITE: "Covalent bonds are broken during boiling." (This is completely incorrect; water vapor still contains intact H–O–H molecules).

CORRECT PHRASING: "Only the **weak intermolecular forces between molecules** are overcome; the strong covalent bonds within molecules remain completely intact."

6. Giant Covalent (Macromolecular) Structures

Extended tier only

💎 Giant Covalent Structures (Macromolecules)

Some non-metal substances do not exist as simple small molecules. Instead, they form **giant three-dimensional networks** of millions of atoms linked together by strong covalent bonds extending throughout the entire structure.

These substances are called **giant covalent substances** or **macromolecules**.

💎 Allotropes of Carbon: Diamond & Graphite

Allotropes are different structural forms of the same element in the same physical state. Carbon exhibits two key allotropes required by Cambridge: **Diamond** and **Graphite**.

1. Diamond

- **Structure:**
 - Each carbon atom is covalently bonded to 4 **other carbon atoms** in a rigid **tetrahedral** arrangement.
 - This regular tetrahedral pattern repeats continuously in three dimensions.
- **Properties & Explanations:**
 - **Extremely Hard:** The rigid 3D framework of millions of strong covalent bonds resists mechanical deformation. It is one of the hardest known natural substances.
 - **Very High Melting Point (approx. 3550°C):** A huge amount of energy is required to break the extensive network of strong covalent bonds.
 - **Electrical Insulator:** All 4 valence electrons of every carbon atom are held tightly in localized covalent bonds. There are **no delocalised electrons** or ions to move and carry charge.
 - **Uses:** Drill tips, cutting tools for glass and stone, gemstone jewellery.

2. Graphite

- **Structure:**
 - Each carbon atom is covalently bonded to 3 **other carbon atoms** in flat, hexagonal rings.
 - This forms **layers (sheets)** of hexagonally arranged carbon atoms.
 - Between the layers, there are **weak intermolecular forces**.
 - The fourth valence electron from each carbon atom is **delocalised** and free to move throughout the layers.
- **Properties & Explanations:**
 - **Soft and Slippery:** The weak forces between layers allow them to easily **slide over one another** when a force is applied.
 - **Good Electrical Conductor:** The **delocalised electrons** are mobile along the layers and can drift when a potential difference is applied, carrying an electrical current.
 - **Very High Melting Point (approx. 3600°C):** Within each layer, carbon atoms are held by very strong covalent bonds requiring vast thermal energy to break.
 - **Uses:** Lubricant for machinery operating at high temperatures, pencil "lead", electrodes for electrolysis.

□ Silicon(IV) Oxide (Silica, SiO₂)

Silicon(IV) oxide is the primary component of quartz and sand.

- **Structure:**
 - Each **silicon atom** is covalently bonded to 4 **oxygen atoms** in a tetrahedral arrangement.
 - Each **oxygen atom** is covalently bonded to 2 **silicon atoms**.
 - The empirical formula is SiO₂ (ratio of 1 silicon atom to 2 oxygen atoms).
- **Comparison to Diamond:**

- Silica has a giant covalent tetrahedral structure very similar to diamond.
- **Properties:** Very hard, extremely high melting point (approx. 1700°C), non-conductor of electricity (all valence electrons localized in bonds).
- **Uses:** Refractory linings for industrial furnaces, glass manufacture.

Comparative Master Table: Diamond, Graphite & SiO_2

Substance	Bonding per Atom	Structure Type	Melting Point	Hardness	Electrical Conductivity	Typical Uses
Diamond	Each C bonded to 4 C atoms	3D tetrahedral network	Extremely high ($> 3500^{\circ}\text{C}$)	Extremely hard	None (insulator)	Cutting tools, drill bits, jewellery
Graphite	Each C bonded to 3 C atoms	2D hexagonal layers with delocalised e^{-}	Extremely high ($> 3600^{\circ}\text{C}$)	Soft & slippery	Good conductor	Electrodes, high-temp lubricant, pencils
Silicon(IV) Oxide	Each Si to 4 O; each O to 2 Si	3D tetrahedral macromolecule	Very high ($> 1700^{\circ}\text{C}$)	Very hard	None (insulator)	Furnace brick linings, sand paper, glass

7. Metallic Bonding & Alloys

Extended tier only

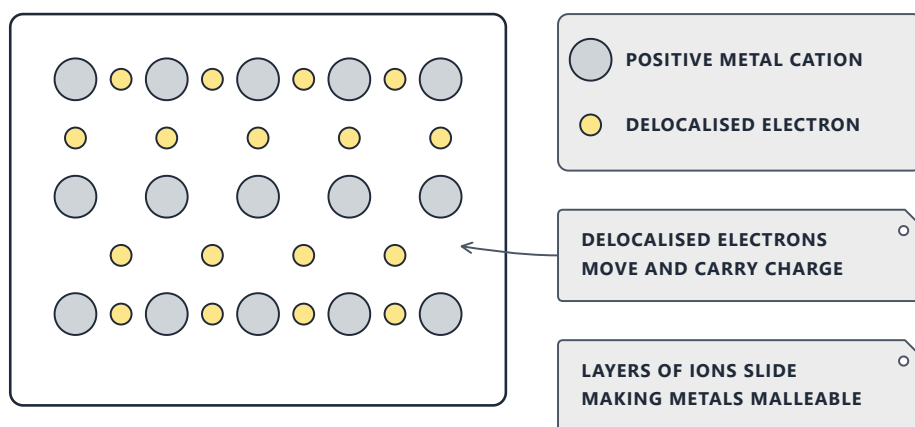
What is Metallic Bonding?

Key Examiner Notes: Cambridge Syllabus Definition

Metallic bonding is the **strong electrostatic attraction** between a **regular lattice of positive metal ions (cations)** and a **"sea" of mobile, delocalised electrons**.

In pure metals:

- Metal atoms lose their outer valence electrons to form a **regular, closely packed lattice of positive cations**.
- The released valence electrons no longer belong to any single atom; they form a mobile **sea of delocalised electrons** that move freely throughout the entire structure.



Explaining the Properties of Metals

Property	Particle Explanation
High Electrical Conductivity	The delocalised electrons are mobile and free to move throughout the structure when a voltage is applied, carrying electrical current.
High Thermal Conductivity	Mobile delocalised electrons rapidly transfer kinetic energy through the lattice; closely packed positive ions also transfer vibrations efficiently.
Malleability & Ductility	In pure metals, the positive ions are arranged in regular layers of identical size . When hammered (malleable) or drawn into wires (ductile), the layers can slide past one another without shattering because the delocalised electron sea moves with them, maintaining the non-directional metallic bond.
High Melting & Boiling Points	The strong electrostatic attractions between the positive metal ions and the delocalised electrons require large amounts of thermal energy to overcome.

What is an Alloy?

An **alloy** is a **mixture of a metal with one or more other elements** (which may be other metals or non-metals like carbon).

Common Examples of Alloys:

- **Brass:** Copper (~ 70%) and Zinc (~ 30%). More corrosion resistant and acoustically resonant than pure copper (used in musical instruments, plumbing fittings).
- **Bronze:** Copper and Tin. Harder and more durable than pure copper (used in statues, medals).
- **Steel:** Iron with small amounts of Carbon (0.2% – 1.5%). Far stronger and harder than pure iron (used in construction girders, bridges, car bodies).
- **Stainless Steel:** Iron mixed with Chromium and Nickel. Resists corrosion and rusting (used in cutlery, surgical instruments).

Why are Alloys Harder and Stronger than Pure Metals?



Key Examiner Notes & Exam Priorities: Core Cambridge Exam Concept (Frequently Examined on Paper 4)

1. In a **pure metal**, all atoms (ions) are of the **same size** and arranged in neat, uniform layers. These regular layers **slide easily** over one another when a force is applied, making pure metals relatively soft.
2. In an **alloy**, the added element has atoms of a **different size**.
3. These different sized atoms **distort and disrupt the regular layered arrangement** of the metal lattice.
4. As a result, it is **much more difficult for the layers of ions to slide past one another**, making the alloy significantly **harder and stronger**.

8. Cambridge Exam-Style Questions & Mark Schemes

Extended tier only · Analisis Langkah Solutif

Contoh Soal & Pembahasan Terpadu

[OSN / IChO Standards]

Question 1 (Paper 2 Multiple Choice — 1 Mark)

An element X has atomic number 12 and mass number 25.
Which statement about an ion of element X is correct?

- **A:** It has 12 protons, 13 neutrons, and 10 electrons.
- **B:** It has 12 protons, 12 neutrons, and 12 electrons.
- **C:** It has 10 protons, 13 neutrons, and 10 electrons.
- **D:** It has 12 protons, 25 neutrons, and 10 electrons.

Examiner Tips and Tricks: Mark Scheme & Step-by-Step Breakdown

Correct Answer: A

Reasoning:

1. Atomic number is 12 $\Rightarrow$ Element X is Magnesium (Mg), which always has 12 protons. (Eliminates C).
2. Neutrons = Mass number – Atomic number = $25 - 12 = 13$ neutrons. (Eliminates B and D).
3. With atomic number 12, its neutral atom has 12 electrons (2, 8, 2). To form a stable ion, it loses 2 valence electrons to achieve an octet (2, 8), carrying a +2 charge and leaving 10 electrons.

Question 2 (Paper 2 Multiple Choice — 1 Mark)

Which substance has a giant covalent structure and conducts electricity?

- **A:** Diamond
- **B:** Graphite
- **C:** Silicon(IV) oxide
- **D:** Sodium chloride

Examiner Tips and Tricks: Mark Scheme & Distractor Analysis

Correct Answer: B (Graphite)

Distractor Analysis:

Diamond (A): Giant covalent, but does **not** conduct electricity (all 4 valence electrons locked in covalent bonds).

Silicon(IV) oxide (C): Giant covalent, but does **not** conduct electricity (insulator).

Sodium chloride (D): Conducts electricity when molten or aqueous, but has a **giant ionic** structure, NOT giant covalent!

Question 3 (Paper 4 Structured Theory — 5 Marks)

Lithium reacts with fluorine to form lithium fluoride, LiF.

(a) Describe, in terms of electrons, what happens when a lithium atom reacts with a fluorine atom.
[3]

> **Mark Scheme:**

- > - **Mark 1:** Lithium loses one electron (to form Li^+).
- > - **Mark 2:** Fluorine gains one electron (to form F^-).
- > - **Mark 3:** Both atoms achieve a full outer electron shell (noble gas configuration / duplet for Li^+ , octet for F^-).

(b) Explain why solid lithium fluoride has a very high melting point. [2]

> **Mark Scheme:**

> - **Mark 1:** Giant ionic lattice with **strong electrostatic attractions** between oppositely charged ions (Li^+ and F^-).

> - **Mark 2:** Requires a **large amount of energy** to overcome/break these strong bonds.

 Question 4 (Paper 4 Structured Theory — 4 Marks)

Pure copper is ductile and can be drawn into electrical wires. Brass is an alloy of copper and zinc.

(a) Explain why pure copper is ductile. [2]

> **Mark Scheme:**

> - **Mark 1:** Copper cations are arranged in **regular layers / rows**.

> - **Mark 2:** Layers of ions can **slide past each other** when a force is applied.

(b) Explain why brass is harder and less ductile than pure copper. [2]

> **Mark Scheme:**

> - **Mark 1:** Zinc atoms are of a **different size** to copper atoms.

> - **Mark 2:** This **disrupts the regular layers**, making it harder for the layers to slide over each other.

Checkpoint Pemahaman & Miskonsepsi

Extended tier only · Latihan Mandiri

SOAL #1 (1. Elements, Compounds & Mixtures)

A student heats a mixture of iron powder and sulfur powder in a test tube. Which observation proves that a chemical compound has formed?

A.

The sulfur melts into a yellow liquid.

B.

A magnet can no longer extract the iron from the product.

C.

The total mass of the test tube and contents remained constant.

D.

The iron powder settles to the bottom of the tube.

Kunci & Penjelasan Konseptual:

In iron(II) sulfide (FeS), the iron atoms are chemically bonded to sulfur atoms. Therefore, iron loses its individual ferromagnetic property and can no longer be separated with a magnet, indicating a compound has formed.

SOAL #2 (2. Atomic Structure & The Periodic Table)

An atom of an element has nucleon number 39 and proton number 19. How many neutrons and what is its electron configuration?

A.

20 neutrons; electron configuration 2, 8, 8, 1

B.

19 neutrons; electron configuration 2, 8, 9

C.

20 neutrons; electron configuration 2, 8, 9

D.

39 neutrons; electron configuration 2, 8, 8, 1

Kunci & Penjelasan Konseptual:

Number of neutrons = Nucleon number (39) - Proton number (19) = 20. With 19 electrons, the shells fill as 2 in shell 1, 8 in shell 2, 8 in shell 3, and 1 in shell 4, giving 2, 8, 8, 1 (Potassium).

SOAL #3 (3. Isotopes & Relative Atomic Mass)

A sample of element X consists of 80% of an isotope with mass number 10 and 20% of an isotope with mass number 11. What is the relative atomic mass (A_r) of element X?

- A.
10.2
- B.
10.5
- C.
10.8
- D.
11.0

Kunci & Penjelasan Konseptual:

$$A_r = ((10 \times 80) + (11 \times 20)) / 100 = (800 + 220) / 100 = 1020 / 100 = 10.2.$$

SOAL #4 (4. Ions & Ionic Bonding)

Why does solid magnesium oxide have a significantly higher melting point (2852 °C) than solid sodium chloride (801 °C)?

- A.
Magnesium oxide contains delocalised electrons.
- B.
The electrostatic attractions between Mg^{2+} and O^{2-} ions are stronger because the charges are 2+ and 2- compared to 1+ and 1- in NaCl.
- C.
Oxygen has a smaller mass number than chlorine.
- D.
Magnesium oxide forms covalent bonds rather than ionic bonds.

Kunci & Penjelasan Konseptual:

Electrostatic force is proportional to the product of charges. In MgO , the ions have double charges (Mg^{2+} and O^{2-}), creating much stronger electrostatic attractions between oppositely charged ions throughout the lattice than the singly charged Na^{+} and Cl^{-} ions in NaCl.

SOAL #5 (5. Simple Molecules & Covalent Bonding)

Why does carbon dioxide (CO₂) have a very low boiling point (-78.5 °C) even though it contains very strong C=O double bonds?

A.

The carbon and oxygen atoms break apart easily when heated.

B.

Only weak intermolecular forces between CO₂ molecules need to be overcome when boiling.

C.

Carbon dioxide is an ionic compound that decomposes at room temperature.

D.

The double bonds in CO₂ repel each other, driving molecules apart.

Kunci & Penjelasan Konseptual:

In simple molecular substances, phase changes overcome the weak intermolecular forces between separate molecules, not the strong covalent bonds within the molecules. Minimal thermal energy is needed to separate the molecules.

SOAL #6 (6. Giant Covalent (Macromolecular) Structures)

Which statement correctly explains why graphite conducts electricity while diamond does not?

A.

Graphite contains mobile positive ions, whereas diamond contains only neutral atoms.

B.

Each carbon in graphite only bonds to 3 others, leaving one delocalised electron per atom free to move along layers.

C.

Graphite has weak intermolecular forces between layers that carry electric current.

D.

Diamond has an open structure allowing air to insulate it against electrical current.

Kunci & Penjelasan Konseptual:

In graphite, each carbon atom uses only 3 of its 4 valence electrons to form covalent bonds with neighbors. The 4th electron is delocalised and free to move throughout the sheet, conducting electricity. In diamond, all 4 valence electrons are locked into localized covalent bonds.

SOAL #7 (7. Metallic Bonding & Alloys)

Why is mild steel (an alloy of iron and carbon) harder than pure iron?

A.

Carbon atoms form covalent cross-links between iron atoms.

B.

Carbon atoms have a different size to iron atoms, disrupting the regular lattice and preventing layers from sliding past each other easily.

C.

Carbon removes all delocalised electrons from iron, forming an ionic crystal.

D.

Carbon atoms are much heavier than iron atoms, increasing the density.

Kunci & Penjelasan Konseptual:

In alloys, atoms of different sizes disrupt the regular rows of the lattice. This prevents the layers of metal ions from sliding smoothly over one another when a mechanical force is applied, rendering the alloy much harder and stronger.